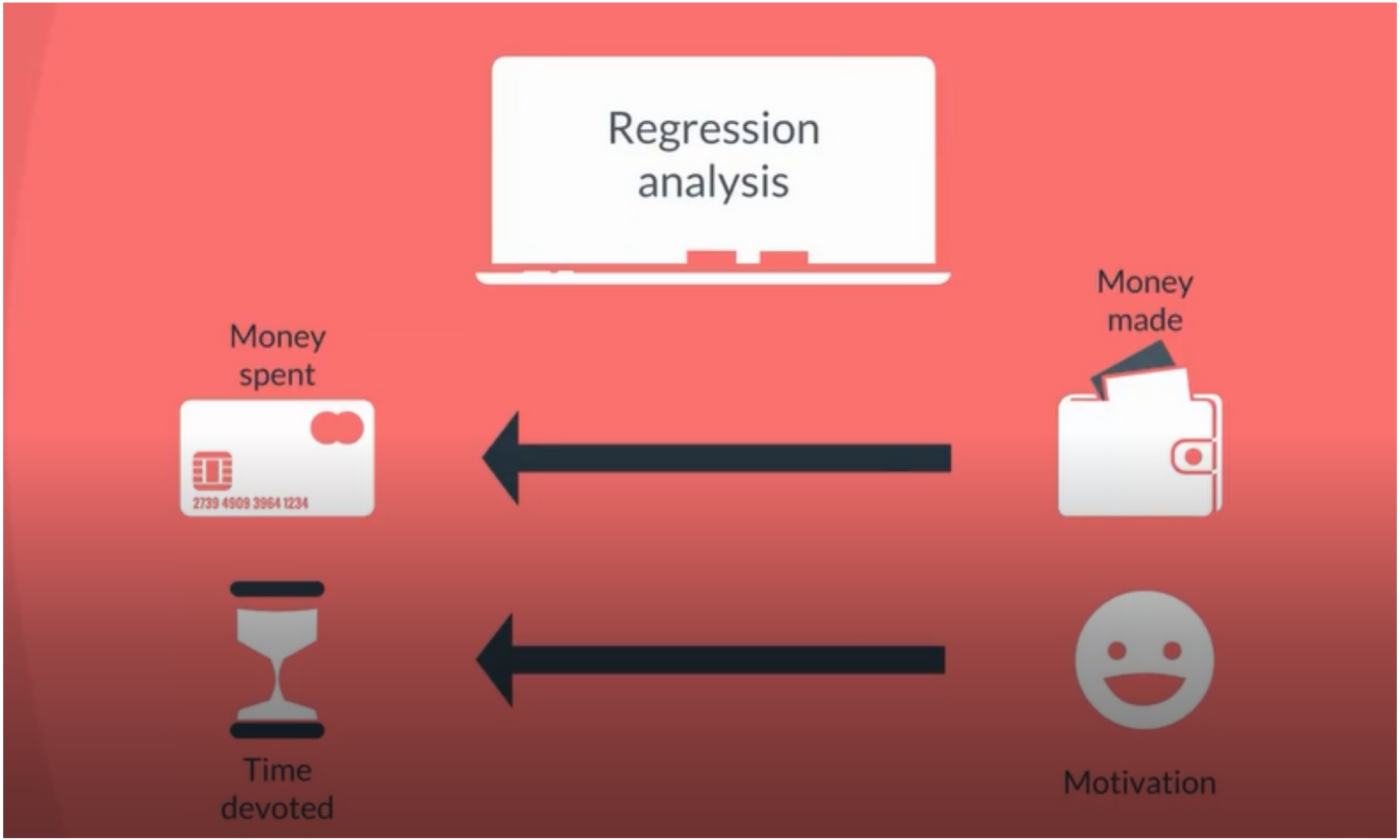

Regression analysis
Money spent
2739 4909 3964 1234
Money made


Regression analysis
Money spent
2739 4909 3964 1234
Money made
Time devoted
Motivation

Systematic approach



Simple linear regression model

How to build a regression

How to interpret a regression

How to compare models

Multivariate linear regression model

After



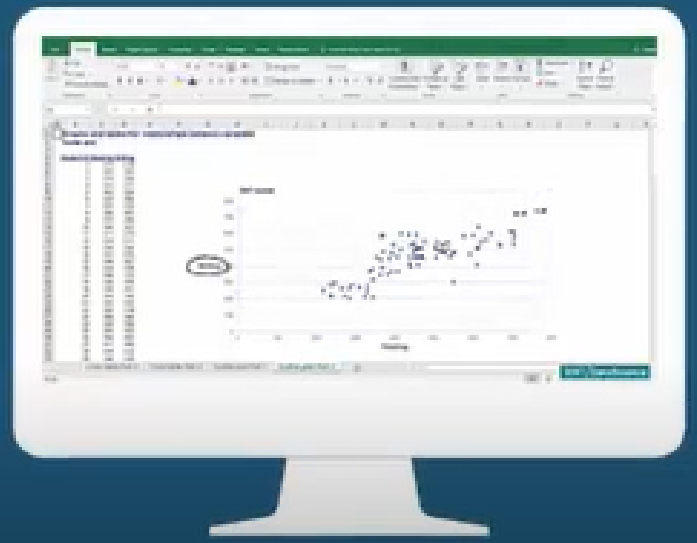
Advanced statistical analysis

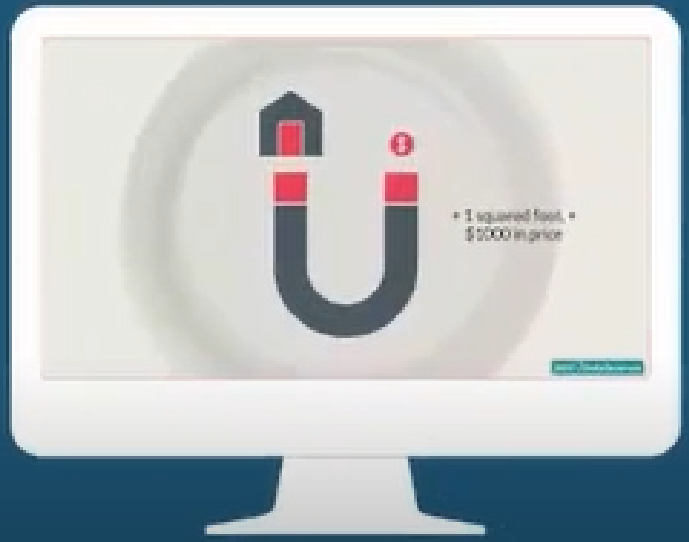


Machine learning



# Recap





# Recap

$$\frac{\sigma_{xy}}{\sigma_x \sigma_y}$$

$$\frac{s_{xy}}{s_x s_y}$$

# Correlation

Vin Diesel example

| | Vin Diesel movies | US GDP growth | UK GDP growth |
|---|---|---|---|
| 2010 | 0 | 2.53% | 1.54% |
| 2011 | 1 | 1.60% | 1.97% |
| 2012 | 3 | 2.22% | 1.18% |
| 2013 | 1 | 1.49% | 2.16% |
| 2014 | 4 | 2.43% | 2.85% |
| 2015 | 4 | 2.43% | 2.33% |

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Correlations

Correlation coefficients are values from -1 to +1, where:
-1 shows very strong negative correlation
0 shows no correlation
1 shows very strong positive correlation

## Correlation

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| Correlations | |
|---|---|
| VD movies - US GDP growth | 0.43 |
| VD movies - UK GDP growth | 0.46 |

# Considerations





Correlation = 0.01

# Sample period 2010-2015





The low GDP growth correlation shows their stronger and weaker times were not coinciding in 2010-2015





0.43





0.01





0.46